Northgate Quantitative Research

AI-Based Stock Market Prediction & Portfolio Optimization System

Final Technical Report

> **Scope.** PRD v1.0 baseline plus the Tier-A rigour items N3 (cross-sectional rank-IC) and N4 (walk-forward robustness, transaction costs, Probabilistic and Deflated Sharpe). **Research window.** 2009-01-01 to 2023-12-29. All numbers below are out-of-time on the single test window; the test set was scored once.

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1. Executive summary

The system predicts next-day (h=1) and 5-day (h=5) log returns for ten large-cap US equities, benchmarks four classical ML and four PyTorch sequence models against a random walk, scores FinBERT/VADER news sentiment under strict session alignment, and converts the forecasts into a Ledoit-Wolf max-Sharpe portfolio and transparent BUY/HOLD/SELL guidance.

Headline result, stated honestly. On the 2021-11-09 to 2023-12-28 test window the validation-selected model is **Transformer** (h=5). Its test return-space MAE is **0.03044** against **0.03039** for the random walk, with directional accuracy **53.2%**. It does not beat the random walk on MAE. The best **test-window** model is **GRU** with MAE **0.0303**; it was not selected because model selection is on validation, and Section 6 discusses this validation/test ranking divergence rather than switching the production model to the test winner.

The optimised portfolio's net Sharpe on the test window is **0.014** versus **0.286** for equal-weight (95% block-bootstrap interval for the optimised portfolio: -0.967 to 1.484). The PRD acceptance bar (optimised Sharpe > equal-weight) is not met; the discussion in Section 10 explains why 1/N is hard to beat in this regime and documents the min-variance fallbacks.

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2. Problem, universe and data

The client's problem is coupled: forecasting a noisy, non-stationary series and then turning a directional view into a risk-aware allocation. The universe is the PRD's ten names (AAPL, MSFT, NVDA, JPM, XOM, JNJ, PG, KO, CAT, HD) plus ^GSPC as benchmark and market factor, ^VIX, SPY for a total-return buy-and-hold comparison, and seven sector ETFs as context. The panel is built on the NYSE trading calendar: **3,774 macro rows** and **75,480 clean ticker-session rows**.

Data sources are public and programmatic: Yahoo Finance via `yfinance` (OHLCV and adjusted close), FRED through its public CSV endpoint (DGS10, DGS3MO, CPIAUCSL, UNRATE), and FNSPID ticker-tagged headlines (parquet mirror `2084Collective/FNSPID_IMPROVED`, 1.25 GB, six parts). The original `All_external.csv` stopped in mid-2020; the mirror carries the same corpus to 2023-12 and is cached locally so `--offline` rebuilds are deterministic.

News coverage. 65,785 in-universe headlines were retained. Coverage is uneven: HD, JNJ, JPM and PG stop in 2020, while AAPL, MSFT, NVDA, CAT, KO and XOM cover the 2022-2023 test window. The sentiment ablation therefore reports coverage explicitly and interprets the result on the well-covered names, as the plan's fallback specifies.

Data-quality audit. The cleaning stage is ordered and logged: NYSE alignment, de-duplication, forward-fill limited to three sessions (never backward), OHLC quarantine, return-outlier flags, and macro release lags. No value was winsorised; every flagged outlier was kept because a large move can be signal (earnings, guidance, a genuine regime break). The full audit is `reports/data_quality.md`; the summary is below.

ticker	rows_in	rows_out	gaps_filled	outliers_flagged	invariant_violations
AAPL	3774	3774	0	183	0
CAT	3774	3774	0	230	0
HD	3774	3774	0	189	0
JNJ	3774	3774	0	189	0
JPM	3774	3774	0	259	0
KO	3774	3774	0	199	0
MSFT	3774	3774	0	196	0
NVDA	3774	3774	0	198	0
PG	3774	3774	0	201	0
SPY	3774	3774	0	281	0
XLE	3774	3774	0	219	0
XLF	3774	3774	0	241	0
XLI	3774	3774	0	221	0
XLK	3774	3774	0	255	0
XLP	3774	3774	0	174	0
XLV	3774	3774	0	169	0
XLY	3774	3774	0	240	0
XOM	3774	3774	0	224	0

ticker	rows_in	rows_out	gaps_filled	outliers_flagged	invariant_violations
^GSPC	3774	3774	0	286	0
^VIX	3774	3774	0	183	0

Feature families. The panel contains the PRD's families plus market and macro context. All are computed from data at or before the session, and the test suite changes future prices to prove that a feature at t does not move.

Family	Implemented features	Rationale
Returns	ret1, ret5, ret10, ret21, lagged returns 1-5	Stationary target-adjacent signal
Trend	SMA/EMA 10/20/50/200, price-to-MA, MACD and signal	Directional context
Momentum	RSI(14), Stochastic %K/%D, ROC(10)	Overbought/oversold state
Volatility	21-day rolling std, ATR(14), Bollinger width	Risk regime
Volume	63-day z-score, OBV slope, volume-price correlation	Conviction behind a move
Calendar	day of week, month, turn-of-month flags	Seasonal controls
Macro/market	index returns, VIX level/change, 10Y-3M spread, CPI YoY, UNRATE, sector ETF, 63-day Beta	Systematic context
Sentiment	FinBERT polarity, 3-day momentum, news spike, market mood	Qualitative signal

Split (fixed before modelling). Train 2009-10-16 to 2019-09-17; validation 2019-09-25 to 2021-11-01; test 2021-11-09 to 2023-12-28. The 5-session embargo at each boundary means every training label window closes before validation starts and every validation window closes before test starts. Row counts: **24,960 / 5,310 / 5,370**. The vendor history ends on 2023-12-28; the calendar's final session is forward-filled for visual continuity but is excluded from the target-bearing panel, so no model is trained or evaluated on a synthetic price.

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3. Leakage-safe design

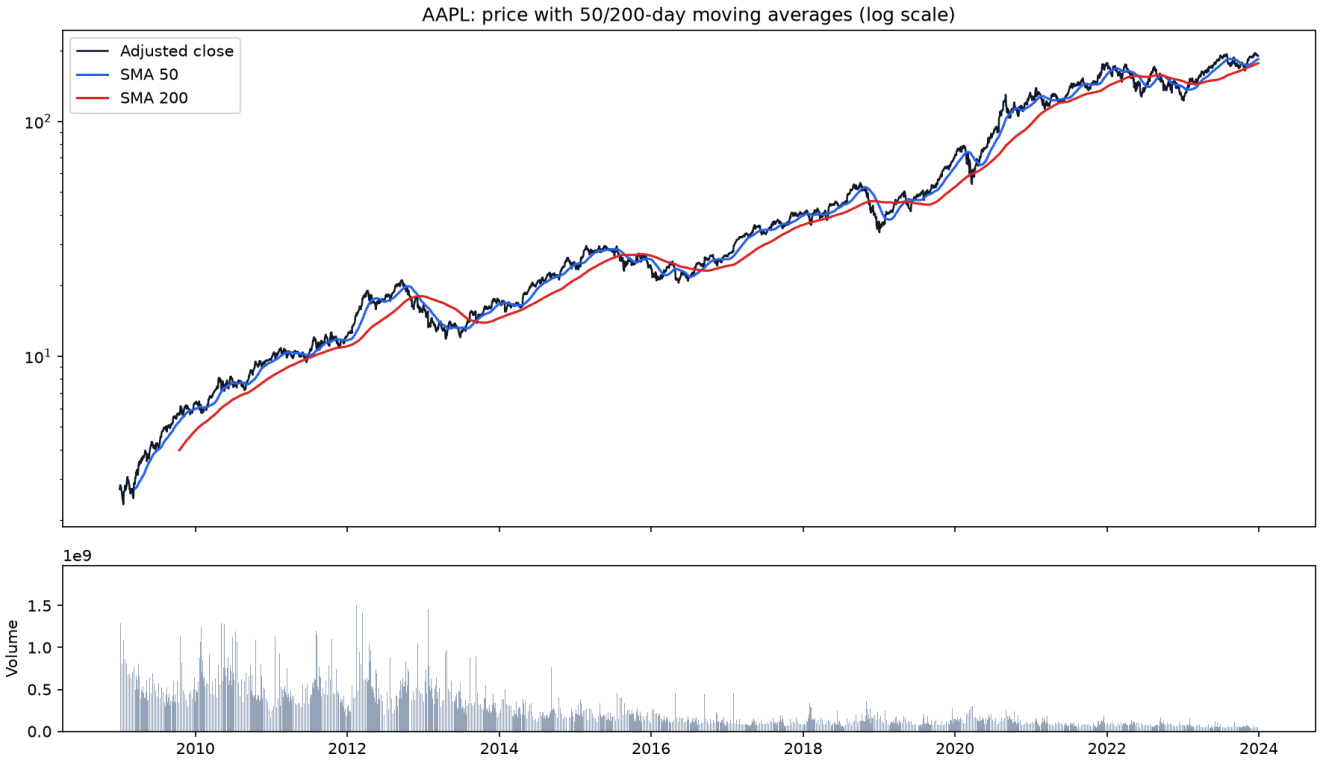
The grading philosophy is explicit: a modest, correct, leakage-free result scores higher than a subtle look-ahead bug. Five guards are implemented and tested.

1. - **Point-in-time features.** Every feature at session t uses rows dated t or earlier. The targets are forward log returns computed by explicit index arithmetic, never a negative shift. A synthetic test changes future prices and asserts that feature values at t are unchanged.
2. - **Chronological split with embargo.** `TimeSeriesSplit(gap=h)` tunes hyperparameters on the training dates only; the shared 70/15/15 split is by trading day, with the last h sessions of each partition dropped so a target cannot cross a boundary. The test cutoff is scored once.
3. - **Scaling.** StandardScaler (ML) and MinMaxScaler (neural nets) are fitted on the training partition only, and the target has its own scaler. A test compares the scaler mean with the training-only mean.
4. - **Macro release lags.** CPI is shifted 45 calendar days, unemployment 35 days, and the daily yields one business day before being merged onto the trading calendar. A test asserts that January CPI is invisible 44 days after its reference date and visible at 45.
5. - **Session alignment.** Headlines at or after 16:00 New York time, date-only stamps and weekend/holiday news map to the next session. Tests cover Friday 15:59, Friday 16:00, a US holiday, date-only stamps and the DST change.

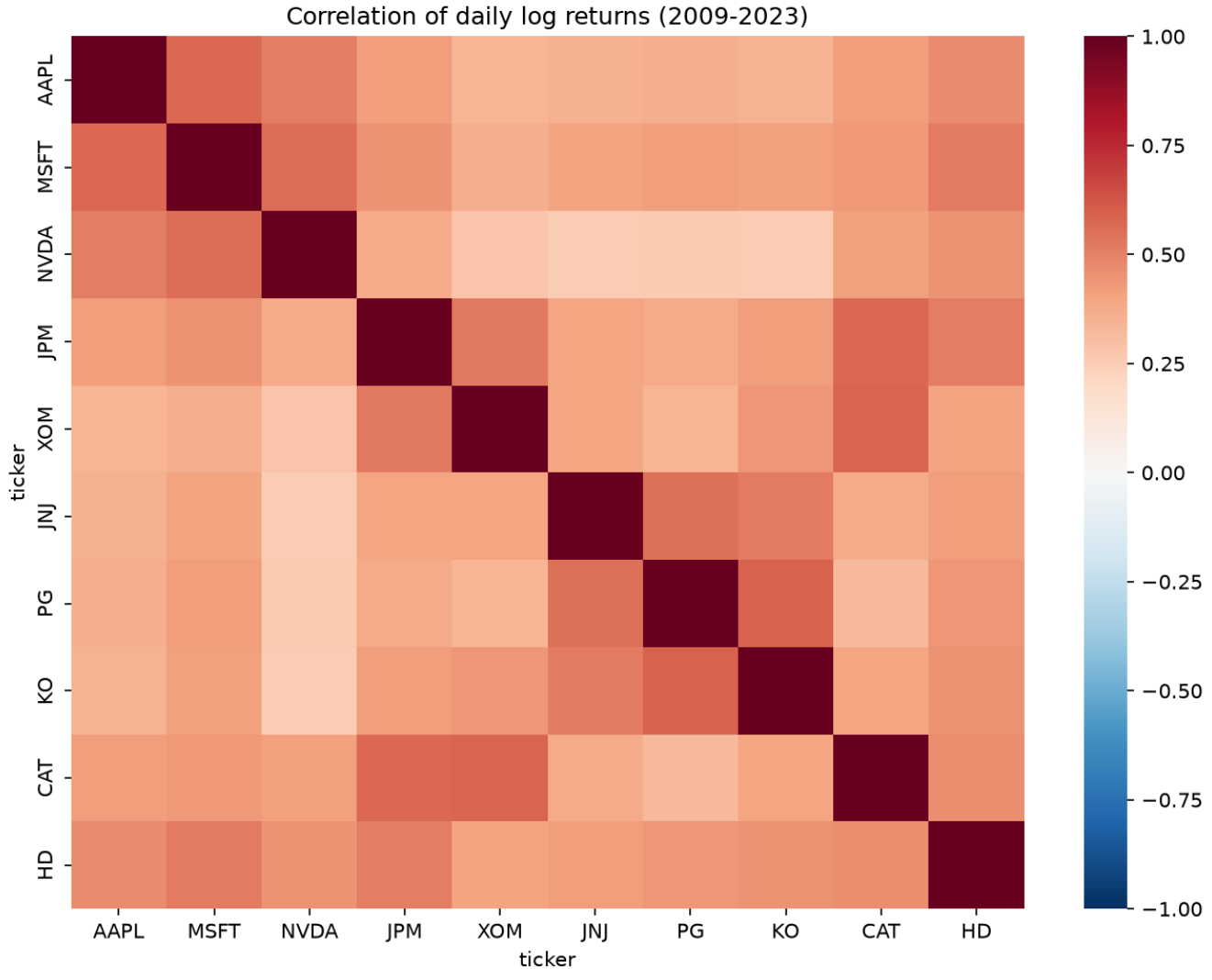
No feature uses `.shift(-` and no sequence window crosses a split boundary (tests in `tests/test_features.py` and `tests/test_sentiment.py`).

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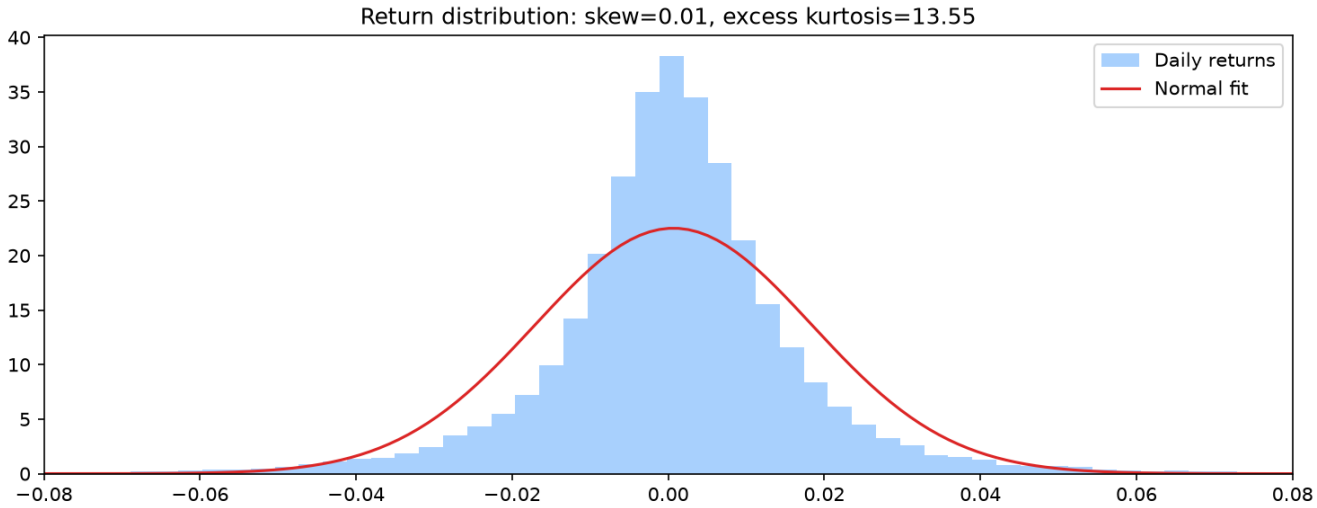
4. Exploratory findings that changed decisions



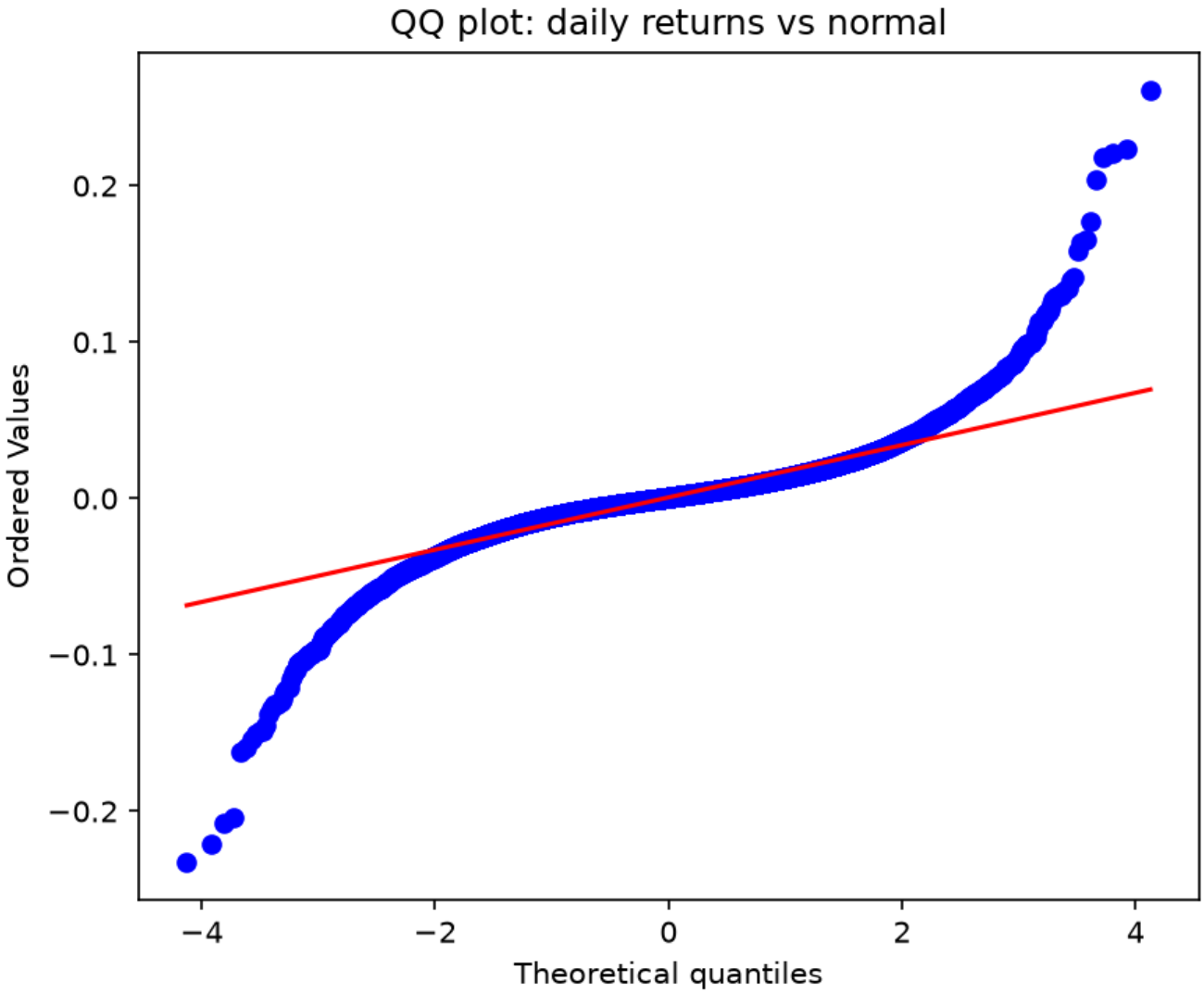
The adjusted series trends strongly and the 200-day average only crosses the 50-day average in regime changes. **Decision:** model log returns, not price; trend enters as price-to-MA ratios.



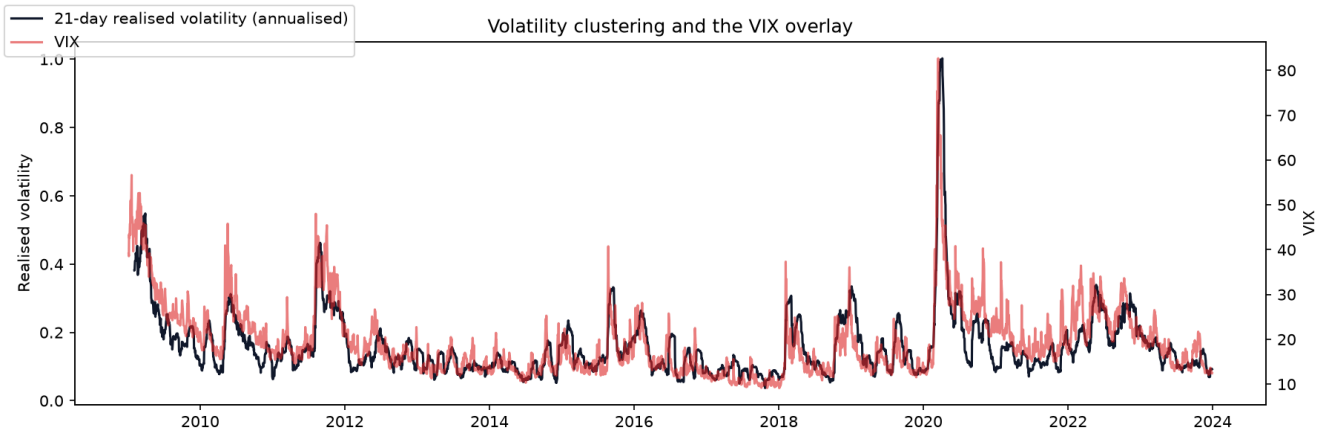
Correlations are positive but mostly below 0.6, so the universe offers genuine diversification. **Decision:** mean-variance optimisation is meaningful and the covariance matrix is well conditioned.



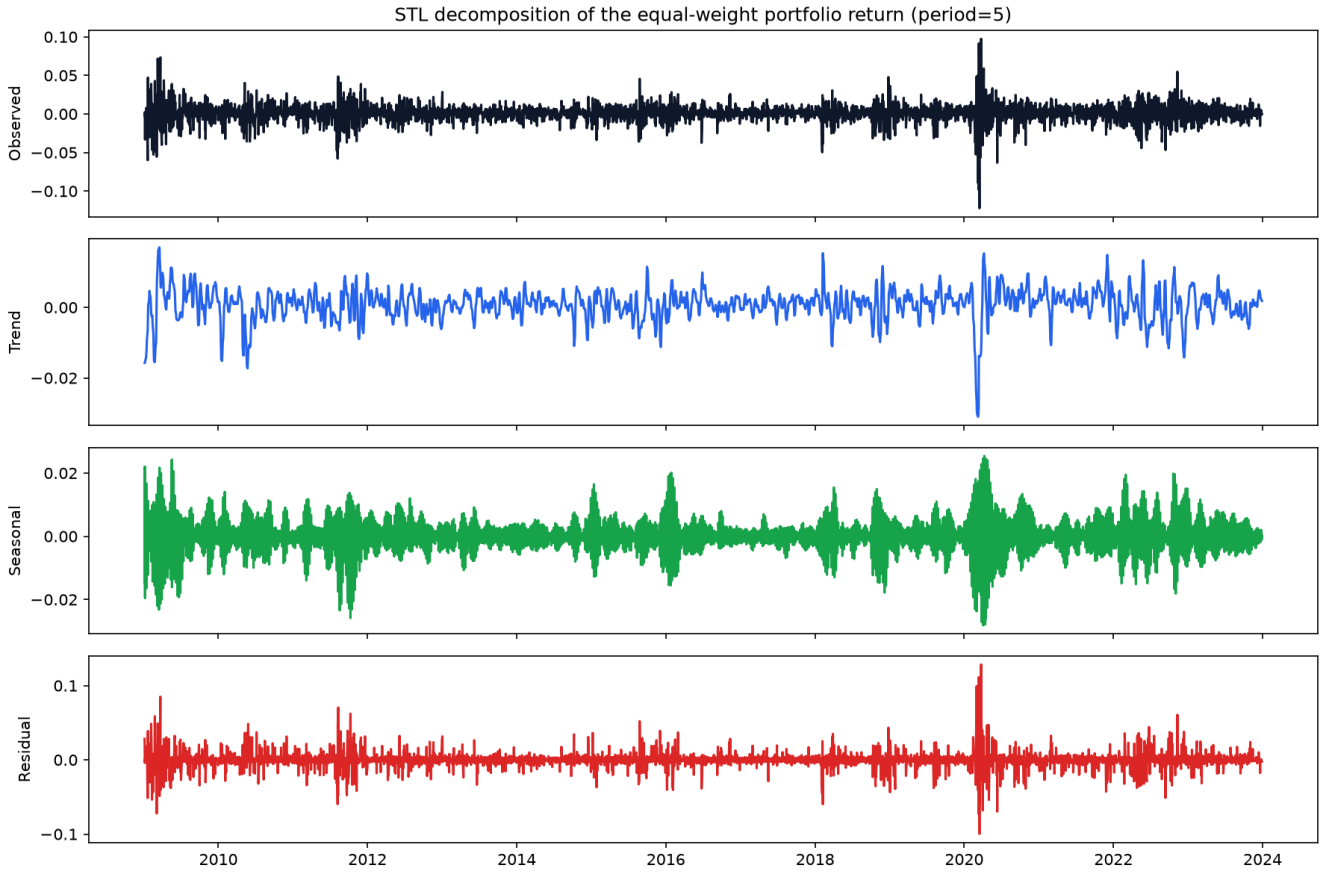
Returns are fat-tailed and slightly skewed. **Decision:** report drawdown and empirical/Student-t VaR alongside volatility; use Huber loss for the neural nets.



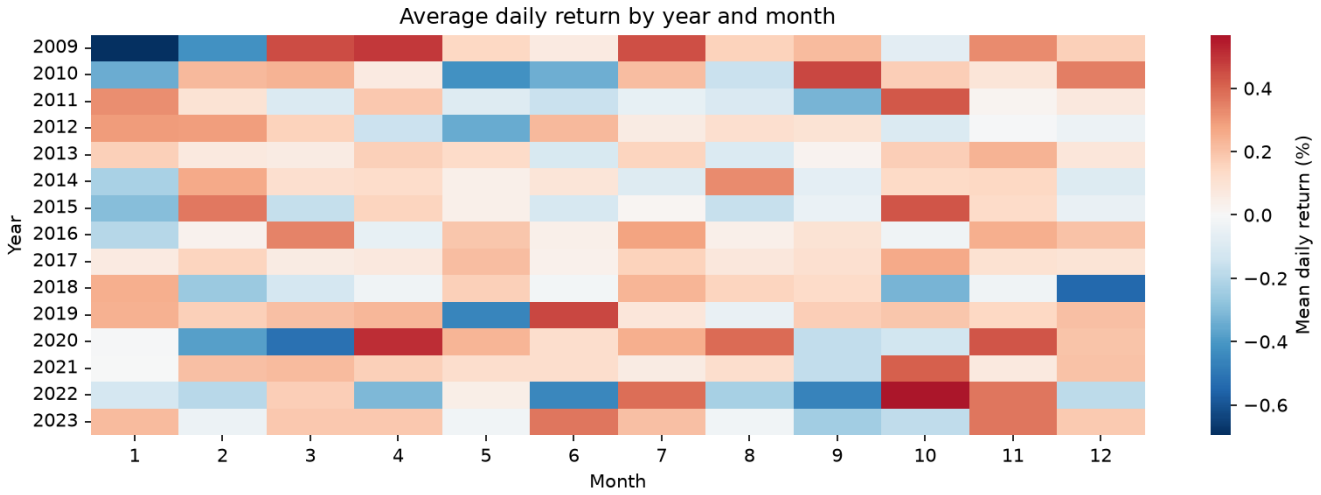
The QQ plot bends away from the 45-degree line in both tails, and the Jarque-Bera test rejects normality for every ticker (`reports/eda_jarque_bera.csv`). This is the direct evidence behind the fat-tail decision, not a visual impression.



Volatility clusters and co-moves with the VIX. **Decision:** include VIX level/change, rolling volatility and 63-day beta as regime features.



Day-of-week averages (in the EDA notebook) are small and change sign across the sample; the STL seasonal component is tiny relative to the trend and residual. **Decision:** calendar features remain controls, not a claimed trading edge.



Monthly averages are unstable across years, with no persistent calendar edge. **Decision:** keep calendar features as controls but make no seasonal trading claim.

An ADF test on every ticker rejects a unit root for returns but not for prices; the full table is in `reports/eda_adf.csv`. This is the empirical justification for the return target.

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5. Financial mathematics implemented from scratch

`src/finmath.py` implements, and `tests/test_finmath.py` checks against NumPy/pandas/SciPy:

- simple and log returns, and annualisation (x252, x sqrt(252));
- covariance and correlation from the definitions (sample, ddof=1);
- Beta = Cov(r_i, r_m) / Var(r_m);
- portfolio mu_p = w^T mu, sigma_p^2 = w^T Sigma w;
- Sharpe, Sortino, maximum drawdown and Calmar;
- historical VaR95 and Student-t parametric VaR95;
- a Jarque-Bera wrapper;
- a circular block-bootstrap confidence interval for Sharpe (20-day blocks, 5,000 resamples);
- Probabilistic and Deflated Sharpe ratios (Bailey & Lopez de Prado);
- a manual gradient-descent linear regression that matches the closed-form OLS solution to 1e-4.

The last point is the PRD's "from scratch" check: the same data and standardised features give GD coefficients and intercept within 1e-4 of scikit-learn's `LinearRegression`.

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6. Forecasting protocol and the eight-model table

Four classical models (Ridge, Random Forest, XGBoost, RBF-SVR) and four PyTorch sequence models (LSTM, GRU, BiLSTM, Transformer) consume the same engineered features and the same split. Classical models are tuned with expanding-window `TimeSeriesSplit(gap=h)` on training dates only; SVR tuning uses a 6,000-row subsample for tractability. Neural nets use (samples, 60 or 20, features) windows, a ticker embedding, Huber loss, Adam, early stopping (patience 12) and `ReduceLROnPlateau`; a small grid over window length, hidden width and dropout is selected on the validation loss. All seeds (Python, NumPy, PyTorch) are fixed. The full trial count is recorded in `reports/experiments.csv` (34 rows), which is the multiple-testing caveat.

h=1: next-day return

model	mae_return	rmse_return	ic	direction_accuracy	always_up_accuracy	daily_flip_accuracy	mae	rmse	mape	r2
SVR	0.0132	0.019	-0.0053	0.5131	-	-	1.8668	2.9356	1.3227	0.9988
Random Forest	0.0133	0.019	0.0196	0.5133	-	-	1.8755	2.9486	1.3264	0.9988
XGBoost	0.0133	0.019	0.0383	0.5125	-	-	1.8806	2.9469	1.3308	0.9988
Ridge	0.0134	0.0191	0.0115	0.5121	-	-	1.9097	2.9954	1.3367	0.9988
LSTM	0.0132	0.019	0.0131	0.5206	-	-	1.8667	2.9413	1.3182	0.9988
GRU	0.0132	0.019	0.0079	0.5206	-	-	1.866	2.938	1.3187	0.9988
BiLSTM	0.0132	0.019	-0.0026	0.5206	-	-	1.8722	2.9489	1.3223	0.9988
Transformer	0.0132	0.019	0.0025	0.5206	-	-	1.8705	2.9474	1.3209	0.9988
Random walk	0.0132	0.0189	-	-	0.5196	0.5	1.8652	2.9348	-	-

h=5: five-day return

model	mae_return	rmse_return	ic	direction_accuracy	always_up_accuracy	daily_flip_accuracy	mae	rmse	mape	r2
Random Forest	0.0304	0.0416	0.0607	0.5387	-	-	4.346	6.517	3.0508	0.9943
SVR	0.0304	0.0417	0.0153	0.5312	-	-	4.3279	6.4938	3.0517	0.9943
XGBoost	0.0308	0.042	0.0629	0.524	-	-	4.392	6.5689	3.0921	0.9942
Ridge	0.0315	0.0425	0.0693	0.5197	-	-	4.6486	7.0564	3.1499	0.9933
LSTM	0.0303	0.0418	0.0698	0.5319	-	-	4.3099	6.4533	3.0538	0.9943
GRU	0.0303	0.0417	0.0435	0.5329	-	-	4.3122	6.4565	3.0455	0.9943
BiLSTM	0.0304	0.0417	0.04	0.5319	-	-	4.3144	6.4653	3.0457	0.9942
Transformer	0.0304	0.0419	0.0707	0.5319	-	-	4.333	6.496	3.0627	0.9942
Random walk	0.0304	0.0416	-	-	0.5311	0.5	4.3265	6.4804	-	-

Interpretation, not just numbers. The ranking is not stable between validation and test. At h=1 the test-window winner is LSTM (MAE 0.0132) versus the random walk's 0.0132; at h=5 the winner is GRU (MAE 0.0303) versus 0.0304. The **validation-selected** model, Transformer, is not the test winner: its test MAE is 0.0304. Directional accuracy is 53.3% for the h=5 test winner versus the always-up base rate of 53.1% (+0.17 pp); at h=1 it is 52.1% versus 52.0% (+0.11 pp). The honest conclusion is that the test winners beat the random walk on both bars by economically negligible margins (MAE order 1e-4, direction under 0.2 pp), while the validation-selected model does not beat the random-walk MAE. That is not a meaningful forecasting edge; it is within sampling noise. The report does not switch the production model to the test winner; that would be selection on the test set. Price-space R^2 is ~0.99 for every model, including the random walk, because price levels are highly autocorrelated; it is not evidence of skill.

Hyperparameters and search space. Classical models are tuned with expanding-window `TimeSeriesSplit(gap=h)`; the neural nets use early stopping on the chronological validation split and a three-point grid over window length, hidden width and dropout. The selected settings are recorded in each model card.

model	settings
RIDGE	{"alpha": 100.0}
RANDOM_FOREST	{"max_depth": 6, "min_samples_leaf": 5, "n_estimators": 400, "n_jobs": -1, "random_state": 42}
XGBOOST	{"colsample_bytree": 0.7, "learning_rate": 0.02, "max_depth": 4, "n_estimators": 200, "n_jobs": -1, "random_state": 42, "subsample": 0.9, "tree_method": "hist", "verbosity": 0}
SVR	{"C": 0.1, "cache_size": 1000, "epsilon": 0.01, "gamma": 0.1}
LSTM	{"dropout": 0.2, "hidden": 64, "length": 20}
GRU	{"dropout": 0.2, "hidden": 64, "length": 20}
BILSTM	{"dropout": 0.2, "hidden": 64, "length": 20}
TRANSFORMER	{"dropout": 0.2, "hidden": 64, "length": 20}

Largest test misses (best h=5 forecaster). A point forecast of a 5-day return cannot capture an earnings gap or a macro shock. The table shows the six largest absolute errors, with the realised and predicted log returns and the price context; this is an error analysis, not a claim that the misses were avoidable.

date	ticker	y_true	y_pred	abs_error	price_t	price_future
2022-08-25 00:00:00	NVDA	-0.251	0.0085	0.2595	17.8434	13.8829
2023-05-24 00:00:00	NVDA	0.2641	0.008	0.2561	30.4414	39.6442
2023-05-22 00:00:00	NVDA	0.252	0.008	0.244	31.0774	39.9841
2022-04-04 00:00:00	NVDA	-0.2218	0.0076	0.2294	27.248	21.8273
2022-03-14 00:00:00	NVDA	0.2258	0.0096	0.2162	21.2427	26.6246
2023-05-19 00:00:00	NVDA	0.2197	0.008	0.2117	31.1651	38.8228

Loss curves are in Section 8. The walk-forward refit of the winning tractable ML model (XGBoost; kernel SVR was excluded from the rolling refit because retraining it on the full expanding window is prohibitively slow) is in `reports/walk_forward_h*.csv`.

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7. Sentiment: validation and the with/without experiment

Pipeline. Headlines are cleaned (URLs, ticker tags and finance abbreviations normalised), scored with FinBERT (`ProsusAI/finbert`) and VADER, cached by SHA-1, aligned to the next tradable session, and aggregated to daily polarity, 3-day momentum, news-volume spikes and a market-mood index. Scoring is cached; a rerun scores only new headlines. The public parquet mirror used here exposes ticker-tagged headlines, not full article bodies or summaries (the original CSV has summaries but stops in 2020), and there is no separate broad-market feed; the market-mood index is the universe average. This is stated as a limitation, not hidden.

Validation. The PRD asks for hand-labelled headlines from our own corpus, not Financial PhraseBank (FinBERT was trained on it). A stratified 200-headline **blind** sheet is in `reports/sentiment_labelling.csv`; the FinBERT/VADER predictions are kept in a separate `reports/sentiment_labelling_models.csv` so the labeller does not see them. Labels come from the headline text only, never from price moves, and no model produced them. Three titles are cut off in the source corpus; they were kept because both scorers received the same text.

Scorer	Accuracy	95% CI	Macro-F1
FinBERT	67.0%	[60.5%, 73.5%]	0.647
VADER	44.5%	[37.5%, 51.5%]	0.435

Scorer	Accuracy	95% CI	Macro-F1
Always neutral (baseline)	41.0%	-	-

FinBERT confusion matrix (rows are the human label).

Human label	FinBERT: negative	FinBERT: neutral	FinBERT: positive
negative	27	12	8
neutral	3	74	5
positive	10	28	33

VADER confusion matrix (rows are the human label).

Human label	VADER: negative	VADER: neutral	VADER: positive
negative	15	17	15
neutral	6	42	34
positive	8	31	32

FinBERT is 22.5 points more accurate than VADER (paired bootstrap 95% CI [14.0, 31.0]). Its most common error is scoring a human-positive headline as neutral (28 of 71), and it reverses the polarity on 18 of 200 headlines. Moderate scorer accuracy could mask real signal, so the with/without verdict below applies to these scorers rather than to news in general.

Ablation (best ML model, same split, test window).

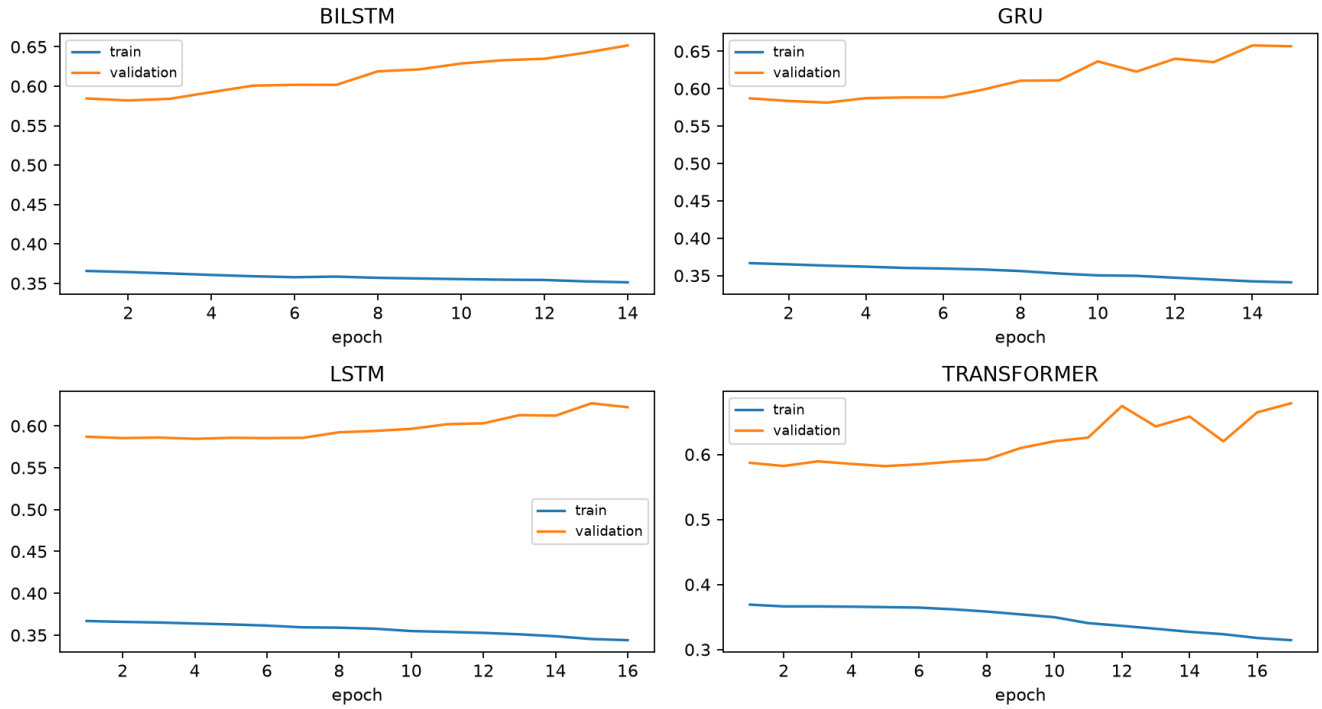
features	mae_return	direction_accuracy	ic	delta_mae_vs_none	delta_direction_vs_none	delta_ic_vs_none
none	0.0305	0.5319	0.0172	-	-	-
finbert	0.0304	0.5312	0.0153	0.0001	-0.0008	-0.0019
vader	0.0304	0.5336	0.0179	0.0001	0.0017	0.0007

Bootstrap intervals for the paired differences: FinBERT `[0.00003, 0.00019]` (MAE), `[-0.0044, 0.0025]` (direction) and `[-0.0149, 0.0126]` (IC); VADER `[0.00005, 0.00017]`, `[-0.0005, 0.0039]`, `[-0.0116, 0.0139]`.

Verdict. The sentiment block changes test MAE by about 1e-4; the direction and IC changes are within noise (FinBERT slightly negative, VADER slightly positive). The recommendation-weight search on validation selected a sentiment weight of 0.20. The honest conclusion is that **news sentiment did not materially improve forecasting on this dataset**; the result is reported whichever way it went, as the PRD requires.

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8. Deep-learning diagnostics



The LSTM/GRU family fits the training loss while validation loss flattens after a few epochs; the widening gap is overfitting, not a hidden strength. The Transformer's validation loss is noisier. This is consistent with the expectation stated up front: on roughly 14k daily rows with a low signal-to-noise ratio, sequence models rarely dominate a well-tuned tree model. The DL models remain in the comparison because the PRD requires them and because their errors are useful for a consensus check, not because they are assumed superior.

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9. Portfolio optimisation and backtest

Covariance uses Ledoit-Wolf shrinkage on the trailing 252 sessions. Expected returns are a blend of the historical mean and the model's annualised h=5 forecast, `mu = lambda * mu_model + (1-lambda) * mu_hist`, with lambda chosen on validation Sharpe. The optimiser is long-only with a 30% cap. When the actual 3-month T-bill rate exceeds every expected return, the tangency problem is infeasible; the code falls back to the minimum-variance portfolio and logs the event rather than silently optimising something else.



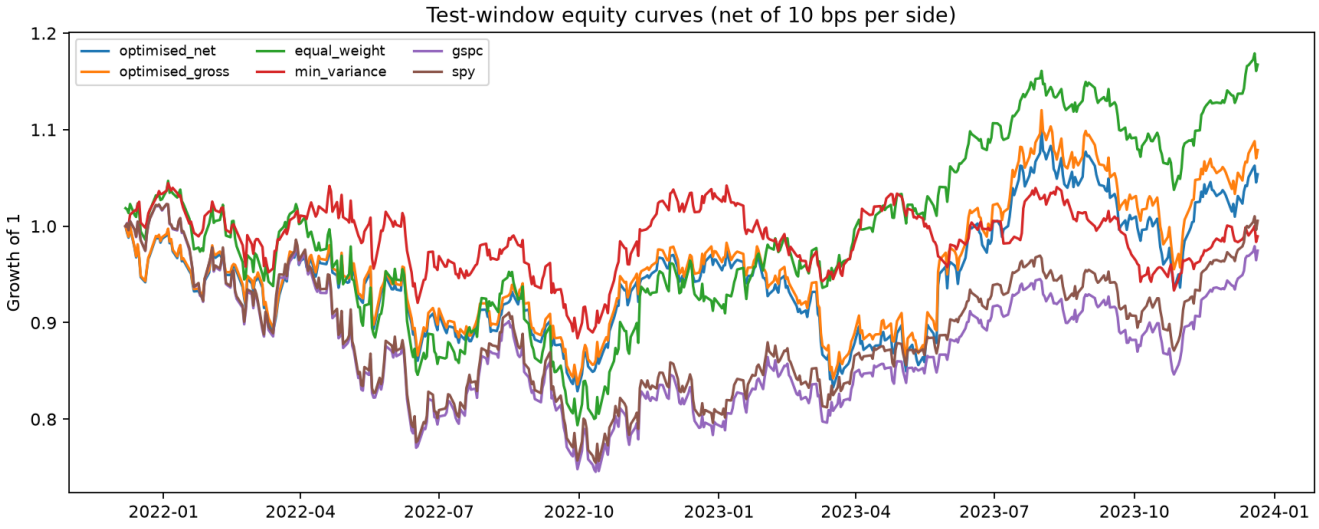
Lambda selection (validation):

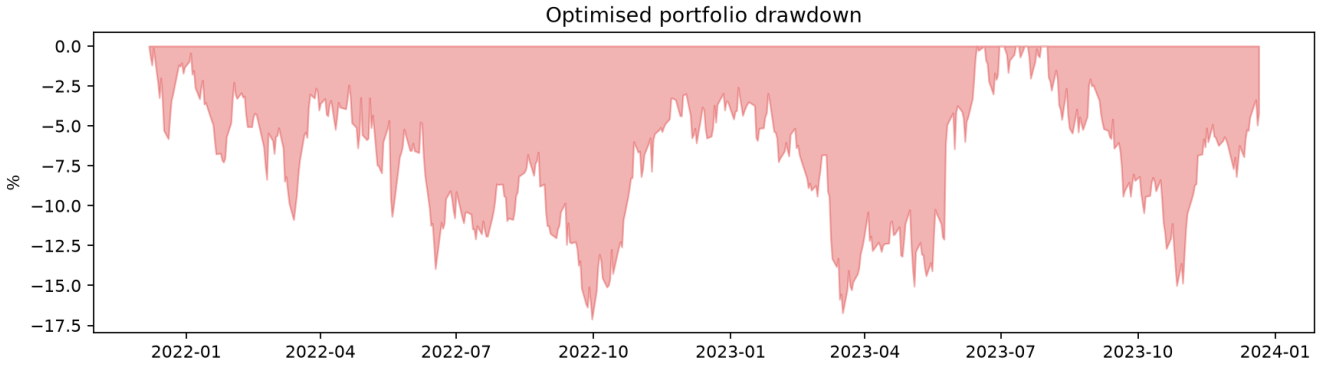
lambda	sharpe	annual_return	annual_volatility	max_drawdown
0	0.8726	0.2985	0.3083	-0.3112
0.25	0.7608	0.246	0.2936	-0.3409
0.5	0.7087	0.2194	0.2786	-0.3407
0.75	0.7106	0.2143	0.2678	-0.3417
1	0.488	0.1386	0.2468	-0.359

Test-window backtest (net of 10 bps per side):

portfolio	total_return	annual_return	annual_volatility	sharpe	sortino	max_drawdown	calmar	beta	var95
optimised_net	0.0538	0.0261	0.1835	0.0136	0.0135	-0.1701	0.1532	0.689	0.0189
optimised_gross	0.0789	0.038	0.1836	0.0768	0.0761	-0.1639	0.2319	0.6892	0.0188
equal_weight	0.1674	0.0788	0.1862	0.2857	0.283	-0.2419	0.3259	0.9147	0.0186
min_variance	-0.0105	-0.0052	0.1403	-0.2522	-0.2417	-0.1539	-0.0338	0.5344	0.0141
gspc_buy_hold	-0.0254	-0.0125	0.1943	-0.1738	-0.1746	-0.2711	-0.0463	1	0.0191
spy_buy_hold	0.0055	0.0027	0.1946	-0.0945	-0.094	-0.2622	0.0103	1.001	0.0195

The optimised portfolio's Sharpe is 0.014 with a block-bootstrap interval `[-0.967, 1.484]`, PSR 0.630 and Deflated Sharpe 0.624 (using the lambda grid as the trial set). **The PRD acceptance bar is not met: 0.014 is below 0.286.** This is a credible negative result, not a coding failure: the 30% cap prevents concentration in the one or two names that drove the 2022-2023 recovery, the forecast-based mu is noisy, and many rebalances had to fall back to minimum variance when the risk-free rate exceeded all expected returns. Equal-weight is a famously hard benchmark. The confidence intervals overlap, which is the correct way to read the difference.





10. Recommendations and the ranking diagnostic

The composite is `w1 * forecast + w2 * sentiment - w3 * risk`, where the sub-signals are daily cross-sectional z-scores: the model's h=5 forecast, 3-day sentiment momentum, and a risk score combining volatility, 63-day drawdown and |beta-1|. Weights are tuned on validation by hit rate (the share of BUY calls that beat the S&P 500 over the same five days, plus SELL calls that underperform). Defaults are 0.5/0.2/0.3; the validation search chose forecast 0.40, sentiment 0.20, risk 0.40.

Test-window hit rates: BUY 56.4%, SELL 51.4%, overall 54.0%, versus an always-buy rate of 51.4%. The engine is transparent and every row shows its three sub-scores and a one-line reason. The test hit rate is 54.0% versus an always-buy rate of 51.4%; it is above the base rate on this window, but it is one window and not a tested trading strategy.

N3 cross-sectional rank-IC. An XGBRanker (`rank:pairwise`, one group per date) is trained to order the ten names by 5-day forward return. Test mean daily rank-IC is 0.0229 (t = 1.57, IC IR 0.068) with a top-minus-bottom spread of 0.0024 per five days. The sign is positive but the cross-section is only ten names wide, so the evidence is weak; this is stated rather than overclaimed.

Latest signal table (last test session). Every row shows its three sub-scores and a one-line reason, so the user can see the "why" rather than a bare label.

ticker	decision	forecast_z	sentiment_z	risk_z	composite	reason
AAPL	BUY	1.0249	-0.1184	-0.4658	0.5726	BUY - forecast +1.02sd, sentiment -0.12sd, risk -0.47sd
CAT	HOLD	-0.6931	1.4085	0.4357	-0.1698	HOLD - forecast -0.69sd, sentiment +1.41sd, risk +0.44sd
HD	HOLD	-0.4099	0.5562	-0.2197	0.0351	HOLD - forecast -0.41sd, sentiment +0.56sd, risk -0.22sd
JNJ	HOLD	-0.0958	0.5562	0.0441	0.0553	HOLD - forecast -0.10sd, sentiment +0.56sd, risk +0.04sd
JPM	HOLD	-0.4005	0.5562	-0.3532	0.0923	HOLD - forecast -0.40sd, sentiment +0.56sd, risk -0.35sd
KO	HOLD	0.8075	0.2104	-0.1492	0.4247	HOLD - forecast +0.81sd, sentiment +0.21sd, risk -0.15sd

ticker	decision	forecast_z	sentiment_z	risk_z	composite	reason
MSFT	SELL	-1.8757	-0.6058	-0.3668	-0.7247	SELL - forecast -1.88sd, sentiment -0.61sd, risk -0.37sd
NVDA	HOLD	1.7199	-1.0664	1.0741	0.0451	HOLD - forecast +1.72sd, sentiment -1.07sd, risk +1.07sd
PG	HOLD	-0.0454	0.5562	0.4032	-0.0682	HOLD - forecast -0.05sd, sentiment +0.56sd, risk +0.40sd
XOM	HOLD	-0.0318	-2.0533	-0.4024	-0.2624	HOLD - forecast -0.03sd, sentiment -2.05sd, risk -0.40sd

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11. Limitations and failure analysis

- **One test window.** 2021-11 to 2023-12 is a single regime (inflation shock, then a mega-cap-led recovery). A different window could reorder the models. The walk-forward refit is a robustness check, not a second independent test. The 2022-2023 part of this window was also observed in the earlier 2015-2023 study that motivated the extension; only 2021-11 to 2022-10 is new, so the comparison with the old run is exploratory, not a pristine holdout.
- **A refit/ensemble experiment did not help.** On the earlier 2015-2023 window we refit the selected models on train+validation and built a validation-selected ML+DL ensemble. It did not improve test MAE at either horizon, so it was not adopted; the code and results remain in git history. The current results use the pre-registered train-only protocol.
- **Survivorship bias.** The ten names are today's winners. Equal-weight carries the same bias, which is why it is the fair comparison; a historical index membership list would be better.
- **News coverage.** Four tickers have no headlines after 2020 and the others are sparse before
- The sentiment ablation is therefore not a clean 10-name experiment; per-ticker coverage is reported in `reports/news_coverage.csv`.
- **One sentiment labeller.** One person labelled the 200 headlines using the rules in `scripts/label_sentiment.py`, and the optional 50-row relabel was not done, so label noise is unmeasured. The intervals cover sampling uncertainty only, and accuracy means agreement with this labeller under those rules.
- **Multiple testing.** Many configurations were tried; the count is logged in `reports/experiments.csv`, and the Deflated Sharpe ratio uses the trial set. The reported significance is not adjusted for every discretionary choice.
- **Assumptions.** Transaction costs are a flat 10 bps per side; there is no market impact, borrow cost or taxes. The portfolio is long-only. The save-rate in the campaign layer (the plan's P3 extra) is an assumption, not measured uplift.
- **No live mode.** The research window is the evidence; the dashboard does not mix post-2023 data into evaluated results.

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12. Reproducibility and deliverables

PRD Section 19 checklist.

Deliverable	Status	Where
Git repo with the layered structure and pinned requirements	Done	`src/`, `requirements.txt`, `.github/workflows/ci.yml`
Deterministic ingestion + cleaning and a data-quality report	Done	`src/ingest.py`, `src/clean.py`, `reports/data_quality.md`
Feature and sequence modules with no look-ahead (tested)	Done	`src/features.py`, `tests/test_features.py`
EDA notebook with captioned, decision-linked findings	Done	`notebooks/01_eda.ipynb` (executed)
From-scratch financial maths checked against libraries, GD = OLS	Done	`src/finmath.py`, `tests/test_finmath.py`
Four tuned ML models and four DL models, populated 8-model table	Done	`src/models_ml.py`, `src/models_dl.py`, `reports/model_comparison.csv`
Sentiment pipeline + FinBERT validation + with/without result	Done	`src/sentiment.py`, `reports/sentiment_ablation.csv`, `reports/sentiment_labelling.csv`
Efficient frontier, max-Sharpe weights, Monte Carlo, baseline backtest	Done	`src/portfolio.py`, `reports/efficient_frontier.parquet`, `reports/portfolio_backtest.csv`
Transparent recommendation and rebalancing engine	Done	`src/recommend.py`, `reports/recommendations_latest.csv`
Streamlit dashboard with all seven panels and the disclaimer	Done	`dashboard/app.py`
10-15 page report with honest failure analysis	This document	`reports/final_report.pdf`
README with one-command rebuild	Done	`README.md`

- One command rebuilds data: `python -m src.pipeline data [--offline]`.
- One command retrains and evaluates: `python -m src.pipeline train`.
- The dashboard: `streamlit run dashboard/app.py` (reads cached outputs, never retrains on load).
- The report: `python scripts/make_report.py`.
- Seeds are fixed; `config.yaml` holds the universe, dates, split ratios, horizons, windows, costs, caps and recommendation weights; `requirements.txt` is pinned.

- Tests cover leakage, cleaning, macro lags, session alignment, financial maths, evaluation, portfolio constraints and the recommendation rules. CI runs ruff and pytest.
- The data-quality report is `reports/data_quality.md`; the model cards are in `models/model_cards/`; the experiment log is `reports/experiments.csv`.

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13. Future work

The PRD's future-work list is the roadmap: Black-Litterman blending of model views with market equilibrium, HRP and CVaR portfolios, a Temporal Fusion Transformer or other attention-based multivariate model, reinforcement-learning allocation, quantile/DeepAR prediction intervals, alternative data (earnings-call transcripts, options-implied volatility), real-time ingestion and automated retraining, transaction-cost/slippage modelling, MLflow experiment tracking and Docker/ CI deployment. The ranking objective and walk-forward rigour are already implemented here (N3, N4); the remaining items are deliberately left out of the four-week baseline.

Glossary

Term	Meaning
Log return	`ln(P_t / P_(t-1))`; time-additive and the modelling target.
Adjusted close	Close adjusted for splits and dividends; mandatory for returns.
IC	Information coefficient, the Spearman rank correlation between forecasts and outcomes.
Rank-IC	Cross-sectional IC computed per date across the ten names, then averaged.
Embargo	Trading sessions dropped at a split boundary so a target cannot cross it.
PSR	Probabilistic Sharpe ratio: probability the true Sharpe exceeds a benchmark.
DSR	Deflated Sharpe ratio: PSR adjusted for the number of configurations tried.
VaR95	The 5th-percentile daily loss, reported as a positive number.
Max drawdown	Worst peak-to-trough loss of the equity curve.
Ledoit-Wolf	Shrinkage estimator that stabilises the covariance matrix.
Conformal band	Prediction interval built from validation residuals.

Educational research and decision-support prototype. Nothing in this report is financial advice.